



A COMMUNITY OF CHOICE

OFFICE OF THE CITY ENGINEER

2101 O'Neil Avenue Room 206

Cheyenne, Wyoming 82001

Phone (307)637-6268 Fax (307)637-6256

MEMO

TO: Mayor Patrick Collins
City Council Members

FROM: Thomas D. Cobb, PE, City Engineer

DATE: October 17, 2022

SUBJECT: Professional Services Agreement between the City of Cheyenne and Terracon Consulting Engineers Proposal for Materials Testing & Special Inspection Services for City of Cheyenne Fire & Rescue Department – New Emergency Response Facilities Converse Avenue, Countryside Avenue, and Hansen Street (3 Locations)

The City of Cheyenne would like to engage a firm that has experience with third party materials testing and specialized inspection services for building and site construction.

I reviewed the Statement of Qualifications on file with the City of Cheyenne and found Terracon with the qualifications, personnel, and expertise necessary to provide the independent professional services required for this important project.

Reason for Agreement:

Third party independent professional specialized inspection and materials testing required for the site and building construction of the City of Cheyenne Fire & Rescue Department – New Emergency Response Facilities located on Converse Avenue, Countryside Avenue, and Hansen Street (3 Locations).

Cost and Funding:

Funding for this project will come from the 2021 1% Special Purpose Sales and Use Tax. The total professional services will be a cost not-to-exceed \$ 121,980. We respectfully request your approval of this professional services agreement.

APPROVED

By John Brodie at 10:44 am, Oct 19, 2022

City Contract No. _____

PROFESSIONAL SERVICES AGREEMENT

for

City of Cheyenne Fire Rescue Department – New Emergency Response Facilities Converse Avenue, Countryside Avenue, and Hansen Street (3 Locations)

THIS AGREEMENT is made and entered into this _____ day of _____, 2022, by and between the City of Cheyenne (“City”), located at 2101 O’Neil Avenue, Cheyenne, Wyoming 82001, and Terracon Consulting Engineers (“Consultant”), 1505 Old Happy Jack Road, Cheyenne, Wyoming 82001.

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the City and the Consultant as follows:

1. **SCOPE OF WORK**

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibit A-1, incorporated by reference, and expressly made a part hereof.

2. **TIME OF PERFORMANCE**

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the scope of work is completed in accordance with Exhibit A-1, unless this Agreement is earlier terminated pursuant to its provisions.

3. **COMPENSATION**

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant a total sum not to exceed one hundred twenty thousand dollars (\$120,000.00).

4. **CONSULTANT RESPONSIBILITY**

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances. Upon notice to the Consultant, the Consultant will, without additional compensation, correct those services not meeting such a standard.

5. **COMPLIANCE WITH LAWS**

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

6. DRUG-FREE WORKPLACE

In compliance with the Drug-Free Workplace Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers, and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees, or sub-consultants are found in violation of this policy, the Agreement may be terminated.

7. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result the performance of the Consultant's duties, including death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

8. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness, or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided

pursuant to an “occurrence” policy. If this coverage is provided pursuant to a “claims made” policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant’s services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said “claims made” coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates “claims made” coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period (“tail coverage”) which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies and shall provide the City with a copy of the endorsements. This requirement does not apply to workers’ compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City’s failure to request or review such policies, endorsements, or certificates shall not

affect the City's rights or Consultant's obligation hereunder. Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

9. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES (49 C.F.R. Part 26)

All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.

10. MONTHLY REPORT

With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.

11. INDEPENDENT CONSULTANT

The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.

12. TAXES

The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts, and obligations incurred by and in connection with its operations under this Agreement.

13. DEFAULT

Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

The term of this Agreement shall be from the date of execution and continue until the scope of work is completed in accordance with Exhibit A-1. The parties may by mutual written agreement renew or extend this Agreement. The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retains all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. OWNERSHIP OF DOCUMENTS AND WORK PRODUCT

All deliverables by the Consultant, all work product of the Consultant, all notes, calculations, memoranda, or any other documentation generated by the Consultant pursuant to the scope of this Agreement, shall be the property of the City of Cheyenne and shall be provided to the City of Cheyenne immediately upon request.

21. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, Wyoming Fair Employment Practices Act of 1965, Wyo. Stat. § 27-9-101, *et seq.*, the Americans With Disabilities Act, as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975, 42 U.S.C. §§ 6101-6107, and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

22. FORCE MAJEURE.

The performance of the Agreement by either party shall be subject to force majeure including, but not limited to, acts of God, fire, flood, natural disaster, war or threat of war, acts or threats of terrorism, civil disorder, unauthorized strikes, governmental regulation or advisory, recognized health threats as determined by the World Health Organization, the Centers for Disease Control, or local government authority or health agencies (including, but not limited to, the health threats of COVID-19, H1N1, or similar infectious diseases), curtailment of transportation facilities, or other similar occurrence beyond the control of the parties, where any of those factors, circumstances, situations, or conditions or similar ones prevent, dissuade, or unreasonably delay the performance required by this Agreement. The Agreement may be cancelled by either party, without liability, damages, fees, or penalty, and any unused deposits or amounts paid shall be refunded, for any one or more of the above reasons, by written notice to the other party.

23. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

24. ENTIRETY OF AGREEMENT

This Agreement consisting of seven (7) pages and Exhibit A-1 consisting of eleven (11) pages, contain the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the parties have hereunto set their hands and seals the day and year first above written.

City of Cheyenne

(SEAL)

By: _____
Patrick J. Collins, Mayor

Attest:

Kristina F. Jones, City Clerk

Date: _____

Consultant

By: _____

Print Name: _____



1505 Old Happy Jack Road
Cheyenne, WY 82001
P (307) 632-9224
Terracon.com

October 13, 2022

City of Cheyenne
2101 O'Neil Ave, Rm 206
Cheyenne, WY 82001

Attn: Mr. Byron Mathews

Re: Proposal for Materials Testing & Special Inspection Services
City of Cheyenne Fire & Rescue – New Emergency Response Facilities
Converse Avenue, Countryside Avenue, and Hansen Street (3 Locations)
Cheyenne, Wyoming
Terracon Proposal No. P24221104

Dear Mr. Mathews:

Terracon Consultants, Inc. (Terracon) appreciates the opportunity to submit this proposal to provide construction materials testing and special inspection services for the **City of Cheyenne Fire & Rescue New Emergency Response Facilities**. Our proposal includes an outline of the project information, our proposed scope of services, estimated quantities, unit rates, and a total estimated fee for our services.

1.0 CAPABILITIES AND EXPERIENCE

Construction Materials Testing & Special Inspection Services

Our team of inspectors and technicians are experienced with providing materials testing, special inspections and/or observations of concrete, soils, aggregate, masonry, structural steel, foundations, fireproofing, and asphalt pavement in the local area and are familiar with the recognized building jurisdiction requirements.

Laboratory Capabilities

Our laboratory is accredited by AASHTO Re:source which is recognized by ASTM E329 Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection. The scope of accreditation includes the field of Soils, Aggregates, Portland Cement Concrete, and Hot Mix Asphalt. As a requirement of accreditation, we regularly participate in the Proficiency Sample Programs of both AASHTO Re:source and the Concrete and Cement Reference Laboratory (CCRL). Our office includes a fully equipped laboratory and employs engineering technicians and special inspectors certified by the American Concrete Institute (ACI), National Institute for Certification in Engineering Technologies (NICET), American Welding Society (AWS), and the International Code Council (ICC). We provide a rigorous internal training program where our staff are evaluated in specific field and laboratory test procedures by internal Terracon auditors and external agencies.

- Accredited by AASHTO Materials Reference Laboratory (AMRL)
- Inspected by Concrete and Cement Reference Laboratory (CCRL)
- Validated by United States Army Corps of Engineers (USACE)

Explore with us

2.0 PROJECT INFORMATION

Our understanding of the required construction materials testing and special inspection services for this project is based upon the following information:

- Project plans by Plan 1 Architects dated 7/22/2022;
- Project specifications titled "Project Manual" by Plan 1 Architects dated 7/22/2022;
- Subsurface Exploration and Geotechnical Engineering Report (GER) titled "Converse Avenue Emergency Response Facility" prepared by Inberg-Miller Engineers dated 5/25/2022;
- Subsurface Exploration and Geotechnical Data Report titled "Hanson Street Fire Hall" prepared by Inberg-Miller Engineers dated 5/31/2022;
- Subsurface Exploration and GER titled "Countryside Avenue Emergency Response Facility" prepared by Inberg-Miller Engineers dated 5/26/2022; and
- Exhibit A – CMTISP Testing Efforts Schedule Unit and Hourly Rate Proposal Sheet – Per site with quantities specified by Plan 1 Architects.

A formal construction schedule was not provided to Terracon prior to the issuance of this proposal, therefore our assumptions on the schedule should be considered an estimate until we can review the contractor's schedule. Pertinent project information is summarized below:

Item	Description
Location	Actual site addresses were not available at the time this proposal was created; therefore, approximate site locations are as follows: ■ Converse Ave. Facility – SW of Converse Ave. and Dell Range Blvd. intersection with approximate coordinates of 41.156886, -104.786100 ■ Hanson Street Fire Hall – West of Hanson St. and Leech Blvd. intersection with approximate coordinates of 41.131910, -104.775119 ■ Countryside Ave. Facility – NW corner of Whitney Rd. and Countryside Ave. intersection with approximate coordinates of 41.153022, -104.733402
Size	Each site encompasses an estimated gross area of approximately 2.75 acres.
Project Description	The proposed site development will be similar at each site location. We understand the projects will include an approximately 11,000 square foot single-story building supported on reinforced concrete footings with reinforced slab-on-grade (SOG), structural steel and light gauge metal framing, CMU structural masonry and brick veneer exterior. The project will also include parking lots, drive lanes, new utilities, concrete sidewalks, curbs, and gutters.
Geotechnical Evaluation	Subsurface Exploration and GER were prepared for each site by Inberg-Miller Engineers as referenced above.
On-site Improvements	Appurtenant site work includes concrete sidewalks, concrete parking areas and drive lanes, curbs, gutters, ADA ramps, a trash enclosure, and other civil site features.

Item	Description
Off-site improvements	Off-site improvements are not anticipated in this scope of work.
Estimated Schedule	Based on conversations with FCI Superintendent Randy Kusler, construction will begin on the Converse Ave. site immediately. He anticipates completion of the materials testing and special inspection scope around March of 2023 for all three projects.

3.0 SCOPE OF SERVICES

Terracon proposes to provide materials testing and special inspection services as summarized below:

Item	Description
Over-excavation & Backfill	Over-excavation of two feet below the bottom of the foundation footings, structural fill import and compaction to be completed per the GERs for each site location. Observations of the excavated material stockpiles and the condition of soils observed at the bottom of the excavations will be performed. We will observe moisture conditioning and backfill operations and perform moisture/density tests on the backfill materials.
Earthwork	Terracon will obtain samples for laboratory tests, perform in-place field density testing and perform periodic or full-time observations for proof rolling, foundation backfill, slab subgrade, select fill, utility backfill, wall backfill, and pavement subgrade and base course.
Foundations	Terracon will observe construction of the reinforced concrete foundations. Terracon will perform the following: ■ Observe the concrete footing construction to verify conformance with the specifications; such as footing width, length and depth, reinforcing steel, plumbness, and condition of the excavated bottom and sidewalls. Other pertinent data can be detailed during the pre-construction specification and review meeting, if requested; ■ Document installation techniques (e.g., ground water, cleaning and concrete placement); ■ Notify the contractor's representative of anomalies, or deviations from the specifications; and ■ Provide an "Open Hole Observation" report signed and stamped by a Wyoming professional engineer.
Concrete Formwork and Reinforcing Steel	Prior to concrete placement, formwork and reinforcing steel for footings, foundation walls, slabs-on-grade and other structural elements will be observed by Terracon. Our observations will include the following: ■ Formwork dimensions; ■ Reinforcing steel size, grade, spacing, cover, position, splices, condition, and supports; and ■ Installation of embeds.

Item	Description
Concrete	During concrete placements, observation, sampling and testing will be performed by Terracon for the concrete used for footings, foundation walls, slab-on-grades, sidewalk, curb and gutter, and parking areas along with other elements. The concrete will be sampled and tested for slump, air content, unit weight, and temperature at the time of placement. Testing will be performed in general accordance with project plans and specifications. ■ Verification of approved design mix; ■ Provide observations of placement procedures, consolidation, and curing and protection; ■ Cast 1 set of 6 (4-inch by 8-inch) cylinders for each 50 cubic yards or fraction thereof; and ■ Compressive strength testing cylinders of 1 @ 7-days 3 @ 28-days and 1 hold cylinder,
Post-Installed Anchors	Terracon will observe and document the post installation of bolts, rebar, threaded rod, and other anchorage systems including observations of materials, hole depth and diameter, cleaning, epoxy preparation, and embedment depth in accordance with contract drawings and the manufacturer's specifications.
Structural Masonry	Terracon will observe the placement of masonry units and reinforcing steel, observe grout spaces, and provide continuous observation during grouting. ■ 1 set of 3 grout compressive strength samples will be obtained for each 5,000 square feet of wall constructed. ■ One set of 3 CMU Prisms or CMU units for compressive strength testing will be obtained for each 5,000 square feet of wall constructed.
Structural Steel	Terracon will perform observations as required for field welded connections, high-strength bolting, and decking. Terracon has assumed that the steel fabrication will be performed by an "approved" fabricator. Fabrication shop observations can be provided if requested. Fabrication shop observations are considered an additional scope of service and will be invoiced on a time and materials basis if needed.
Asphalt Concrete Pavement	Relative field density-compaction testing by nuclear methods will be performed during asphalt concrete paving operations. The density-compaction of the asphalt concrete will be evaluated utilizing information from the paving contractor's asphalt mix design. If requested, asphalt samples and cores will be obtained during each day of asphalt paving operations and tested for unit weight, density, and asphalt content and gradation.
Cold Formed Steel Framing	Terracon will observe the cold form steel framing according to the requirements of the specifications: ■ Documentation of certified welders for joint configuration and welding process. ■ Member size, location and spacing. ■ Bracing and stiffeners. ■ Joint details at connections. ■ Screwed, bolted, welded and power-actuated fasteners utilized.

Item	Description
Project Management	A project manager will be assigned to the project to review the daily activity and assist in scheduling the work. Field and laboratory tests will be reviewed prior to final submittal. The project manager will be responsible for the project budget, communicating with the contractor regarding schedule, deviations, and documenting the resolution of outstanding deviations. To help create a good working relationship with the contractor and for the contractor to better understand our scope of work for the project, we request that Terracon be invited to preconstruction meetings prior to each phase of construction.

Commitment to Timely Report Turnaround:

We understand the importance of report turnaround to our clients and we are committed to delivering test results on a timely basis as well as the following reporting standards:

- Failing tests or non-conformance items will be communicated to the designated parties prior to leaving the site and handwritten draft reports are available at the end of each day.
- Final reports with non-conformances (Deviation Reports) will be provided within 24 hours of testing.
- Final reports without non-conformances will be provided within five business days.
- Final laboratory test reports will be provided within two days of test completion.

Terracon continuously applies new processes, methodologies, and techniques to solve project challenges cost-effectively, especially when it comes to timely delivery of high-quality field data reports so our clients can ***make quick, informed decisions much sooner than ever before.***

Our proprietary data management systems generate real-time results of critical information, using data captured in the field via mobile applications, keeping our clients informed while simultaneously moving the project forward.

Terracon's seamless constructed delivery approach is used every day when we provide materials services for tens of thousands of projects. This innovative approach was created to ultimately benefit our clients in receiving higher quality, more consistent reports and deliverables even faster, all with the flexibility to provide this information when and where you and the project team need it.

- Captured Global Positioning System (GPS) test locations and descriptive diagrams allow non-conformance items and failing tests to be resolved more quickly.
- Field data collection mobile applications used to power form-based and map-based workflows, which all users to complete digital forms anywhere, anytime, even offline
- Customization features and high-quality, consistent reports including test locations on a site plan and photographs that can be shared with multiple entities.

SEAMLESS CONSTRUCTED DELIVERY



- Real-time reporting through Terracon's various client-facing platforms keeps **the** project team informed, ***saving time and money.***

Through a combination of Terracon tools powered by mobile field data collection applications, we make it easy for our clients to access project data, documents, and reports all in one location through a collaborative online experience.

These tools allow any Terracon user to visually place field observations and tests right where they belong on your plan, in real time, entering data throughout the day through mobile devices. When used for field data collection and reporting – our innovative tool system gives **our clients** a ***huge technological advantage in project communication and collaboration through visual, map-based data review and analysis.***

Additional benefits of our seamless constructed delivery, enable Terracon to strive for on-time project delivery to our clients in the following ways:

- **Project Work Plan and Schedule**
 - Experienced staff with the right certifications
 - Tracking and responding to manpower needs
 - Responding to workload fluctuations
- **Budget Tracking**
 - Constant communication – daily, weekly, monthly
 - Milestone notifications monitor your investment
 - Retest tracking for potentially significant savings
- **Field Observation, Data Collection, Reporting and Delivery**
 - Fast collection through hand-held mobile devices, high-tech tablets, and laptops
 - Field representatives have critical project design information and specifications at their fingertips and readily accessible
 - Efficient transmittal of data from the field, to the cloud, to the project manager; what used to take hours now takes seconds
 - Keeps the project moving forward with seamless delivery of information
- **Communication of Results to All Parties**
 - E-mail notifications
 - Easy to read reports with third-party access
 - Greater consistency through standardized report interface – eliminates reporting "gaps" or missing information
 - Close out – push of a button
 - Report library
 - Data storage
 - Final letters

Terracon Field Representative: In addition to the above services, our field personnel will provide the following services during their visit:

- Check in with the project general superintendent upon arrival on-site.
- Confirm that current approved construction documents are available during our visit.
- Notify the general contractor of our field observations and test results prior to leaving the site.
- Submit a written draft report to the Terracon project manager for review.

- Our field personnel have the right to decline work if they believe the conditions are not safe.

Scheduling of Services: We understand that the client may not be involved with scheduling our services; this is typically the responsibility of the general contractor. We request that the following information be passed on to whom will be responsible for scheduling our services.

- Scheduling testing services must be requested no later than 3:00 pm on the business day preceding the work.
- For structural steel testing and observations, we request 72 hours' notice.
- Scheduling is performed through our dispatcher by calling (307) 632-9224.
- Cancellation of services should be done prior to a Terracon representative mobilizing to the project. Failure to do so will result in a cancellation fee.
- Terracon will not be responsible for tests that are not performed due to a failure to schedule our services on the project.
- Testing and observations will only determine compliance with project specifications at the test locations, at the time our services are performed.
- The geotechnical report was not provided in the preparation of this proposal. Once this information becomes available, we request to review and revise our proposal accordingly.

4.0 COMPENSATION

Based on the project information available for our review and the quantities provided in Exhibit A – CMTISP Testing Efforts Schedule Unit and Hourly Rate Proposal Sheet – Per Site, our estimated budget to perform the proposed scope of services is ***\$40,660 per site with a combined fee of \$121,980 for all three sites***. Fees for services provided will be based on the unit rates shown in that exhibit. Please note that this is only a budget estimate and not a not-to-exceed price. Any additional services not specified in the provided CMTIP Fee Schedule or out of scope items will be performed within accordance with our attached Material Service Line Fee Schedule.

Many factors, including those out of our control, such as weather and the contractor's schedule including overtime and weekend work, and the need for re-testing will dictate the final fee for our services. We will not exceed our budget without first notifying you and providing a summary of work performed to date and remaining work. We will track the costs of re-testing, stand-by time, and cancellations separately.

For purposes of our proposal, overtime is defined as all hours worked above 8 hours per day, between the hours of 5:00pm and 7:00am, Saturday, Sunday and Terracon recognized holidays. Overtime rates will be 1.5 times the hourly rate quoted. All charges are portal to portal from our local office. A minimum of 2 hours will be charged for each site visit with the exception of sample pick-ups.

5.0 ASSUMPTIONS

- We understand that there are no retaining walls planned for construction during this phase of construction;
- We have assumed that contractors on the site will work a single shift, typical schedule of 5 days per week;
- An out of scope charge at the applicable unit rate will be applied for all stand-by time and/or time spent on activities which are not cancelled with prior notice;

- Staff time, laboratory testing and extra trips required for re-testing and re-inspections will be denoted on the monthly invoices as Task 99 and be considered out-of-scope items not included in the above estimated budget.
- A complete construction schedule was not provided to Terracon prior to the issuance of this proposal, therefore our assumptions on the schedule should be considered an estimate until we can review the contractor's schedule.

6.0 AUTHORIZATION

This proposal may be accepted by providing Terracon with a City of Cheyenne agreement for review and execution. Please be aware that we will be unable to distribute field and laboratory reports until a signed contract is received. This proposal is valid only if authorized within 45 days from the proposal date.

We appreciate the opportunity to provide this proposal and look forward to working with you. Please give us a call if you have any questions or comments regarding this proposal.

Sincerely,
TERRACON CONSULTANTS, INC.



Zach St. Jean, P.E, P.G.
Office Manager / Principal



Kyle Royse
Materials Department Manager

Copies to: Addressee (via e-mail)

Enclosures: Exhibit A – CMTISP Schedule (provided by City of Cheyenne)
Material Service Line Fee Schedule

EXHIBIT A

CMTISP Testing Efforts Schedule Unit and Hourly Rate Proposal Sheet - Per Site

A	B	C	D	E	F	G
Bid Item	TEST TYPE [Provided by Architect]	GOVERNING SPECIFICATION SECTION [Provided by Architect]	UNIT [Provided By Architect]	ESTIMATED QUANTITY OF TEST [Provided by Architect]	UNIT COST PER TEST [Provided by CMTISP]	PROPOSED FEE FOR TEST [E X F]
1	Concrete Footer, Wall, and Slab Inspection	03 30 00	Visits	8	\$ 330	\$ 2,640.00
2	Concrete Footer, Wall, and Slab Concrete Tests	03 30 00 Sect 3.16D	1 per 50 CY of concrete placed per day	12	\$ 125	\$ 1,500.00
3	Site Concrete Tests	03 33 05 Sect 3.2	1 per 20 c y single pour or 1 per 50 c y accumulative	30	\$ 125	\$ 3,750.00
4	CMU Prism Test, Mortar Test, Grout Test	04 22 00	1 per 5000 sf of wall area	4	\$ 540	\$ 2,160.00
5	CMU Wall Inspection	04 22 00	Visits	4	\$ 330	\$ 1,320.00
6	Structural Steel & Steel Joists Inspection	05 12 00	Visits	3	\$ 535	\$ 1,605.00
7	Steel Decking Inspection	05 31 00	Visits	2	\$ 535	\$ 1,070.00
8	Cold-Formed Steel Framing Inspections	05 40 00	Visits	2	\$ 535	\$ 1,070.00
9	Proctor - Native Material	31 23 16	Each	1	\$ 160	\$ 160.00
10	Gradation - Native Material	31 23 16	Each	1	\$ 275	\$ 275.00
11	Field Density - Subgrade in Paved Areas	31 23 16 Sect 3.2	1 per 500 s y	10	\$ 220	\$ 2,200.00
12	Field Density - Subgrade for Curb & Gutters	31 23 16 Sect 3.2	1 per 300 l f	4	\$ 220	\$ 880.00
13	Field Density - Subgrade for Sidewalks	31 23 16 Sect 3.2	1 per 1,200 s f	5	\$ 220	\$ 1,100.00
14	Proof Roll - Subgrade in Paved Areas	31 23 16 Sect 3.3	Each	1	\$ 220	\$ 220.00
15	Field Density - Trench Backfill	31 23 16 14 Sect 3.3	1 per 100 l f for each 16-inch level	12	\$ 220	\$ 2,640.00
16	Field Density - Utility Manholes & Inlets	31 23 16 14 Sect 3.3	2 per 16-inch level	15	\$ 220	\$ 3,300.00
17	Proof Roll - Trench Backfill in Paved Areas	31 23 16 14 Sect 3.3	1 per trench	2	\$ 220	\$ 440.00
18	Proctor - Crushed Base	32 11 20	Each	1	\$ 160	\$ 160.00
19	Gradation - Crushed Base	32 11 20	Each	1	\$ 275	\$ 275.00
20	Field Density - Crushed Base in Paved Areas	32 11 23 Sect 3.3	1 per 500 s y	10	\$ 220	\$ 2,200.00
21	Field Density - Crushed Base for Curb & Gutters	32 11 23 Sect 3.3	1 per 300 l f	4	\$ 220	\$ 880.00
22	Field Density - Crushed Base for Sidewalks	32 11 23 Sect 3.3	1 per 1,200 s f	5	\$ 220	\$ 1,100.00
23	Proof Roll - Crushed Base in Paved Areas	32 11 23 Sect 3.3	Each	1	\$ 220	\$ 220.00
24	HMA Volumetrics (Marshall, Gradation, Asphalt Content)	32 12 16 Sect 3.4	1 per 1,000 tons	1	\$ 610	\$ 610
25	Field Density - Asphalt Pavement Sample Retrieval - Concrete,	32 12 16 Sect 3.4	1 per 500 tons	1	\$ 430	\$ 430
26	Masonry, Soil, Crushed Base		1 per specimen	15	\$ 150	\$ 2,250

Estimate 4 hours per inspection @ \$80/hr. and 10 miles
roundtrip @ \$1 00/mile

Estimate 5 cylinders/set @ \$25/cylinder

Estimate 5 cylinders/set @ \$25/cylinder
Estimate one set of (3) masonry assemblages/5000 sf of
wall @ \$180/unitEstimate 4 hours per inspection @ \$80/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 5 hours per inspection @ \$105/hr. and 10
miles roundtrip @ \$1 00/mileEstimate 5 hours per inspection @ \$105/hr. and 10
miles roundtrip @ \$1 00/mileEstimate 5 hours per inspection @ \$105/hr. and 10
miles roundtrip @ \$1 00/mile

ASTM D698 standard proctor

Standard Sieves and Liquid-Plastic Limits

Estimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mile

ASTM D698 standard proctor

Standard Sieves and Liquid-Plastic Limits

Estimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 3 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mile

Theoretical Max Specific Gravity

Estimate 6 hours per inspection @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mileEstimate 2 hours per retrieval @ \$70/hr. and 10 miles
roundtrip @ \$1 00/mile

EXHIBIT A

CMTISP Testing Efforts Schedule Unit and Hourly Rate Proposal Sheet - Per Site

A	B	C	D	E	F	G
Bid Item	TEST TYPE [Provided by Architect]	GOVERNING SPECIFICATION SECTION [Provided by Architect]	UNIT [Provided By Architect]	ESTIMATED QUANTITY OF TEST [Provided by Architect]	UNIT COST PER TEST [Provided by CMTISP]	PROPOSED FEE FOR TEST [E X F]
27	Report Review - Inspections		1 per site visit	89	\$ 45	\$ 4,005
28	Report Review - Lab Tests		1 per lab test	5	\$ 45	\$ 225
29	Project Management		As needed	10	\$ 135	\$ 1,350
30	Open Hole Report		Per Report	1	\$ 250	\$ 250.00
31	Project Geologist		Per Hour	3	\$ 125	\$ 375.00
					\$ -	\$ -
PROPOSED UNIT PRICE FEE NOT TO EXCEED						\$ 40,660.00

Review and distribution of field reports

Review and distribution of laboratory reports

Site visits, consulting, safety, project coordination

Open Hole Report

Site visit to perform Open Hole Inspection

Hourly Fees for Field Observation of the Work

A	B	C	D	E	F	G
Bid Item	Description of Hourly Services [Provided By Architect]	STANDARD HOURLY RATE [Provided By CMTISP]	OVERTIME HOURLY RATE [Provided By CMTISP]	ESTIMATED STANDARD HOURS By CMTISP	ESTIMATED OVERTIME HOURS [Provided By CMTISP]	PROPOSED FEE FOR OBSERVATION HOURS ((C X E) + (D X F))
1		\$ -	\$ -	0	0	\$ -
2		\$ -	\$ -	0	0	\$ -
3		\$ -	\$ -	0	0	\$ -
4		\$ -	\$ -	0	0	\$ -
5		\$ -	\$ -	0	0	\$ -
PROPOSED OBSERVATION FEE NOT TO EXCEED						\$ -

Proposal Summary

PROPOSED UNIT PRICE FEE NOT TO EXCEED	\$ 40,660.00
PROPOSED OBSERVATION FEE NOT TO EXCEED	\$ -
TOTAL FEE PROPOSAL NOT TO EXCEED	\$ 40,660.00

CMTISP provider warrants that unit prices and hourly fees submitted are inclusive all of subsistence per diem, mileage, clerical time, sample pickup, processing time, etc. to complete and provide the testing services specified in the agreement.

Construction Material Testing and Inspection Service Provider's Signature  Date **10/13/2022**

TERRACON CONSULTANTS, INC.

MATERIALS SERVICE LINE FEE SCHEDULE

PROFESSIONAL SERVICES

TECHNICIANS AND CLERICAL	Hourly Rate
Sr. Field/Lab Technician (State DOT or Equivalent Certification)	\$75
Field or Laboratory Technician	70
Floor Flatness/Levelness Technician	115
Overtime surcharge (before 7 a.m., after 5 p.m. M-F, Saturdays, Sundays, and Holidays)	1.5 x's hourly rate
Special Inspection (Masonry, Reinforced Concrete, Fireproofing)	80
Structural Steel-Weld-Bolted Connection	105
Clerical/Dispatch	65

ENGINEERING	Hourly Rate
Principal (Sr. Engineer/Geologist)	\$180
Senior Project Engineer/Manager	160
Project/Department Manager	135
Project Geotechnical Engineer / Geologist	125
Staff Engineer/ Geologist	115
Field Engineer / Geologist	100
Special Consultation, Expert Testimony, & Court Appearance	185-300

FIELD AND LABORATORY TESTING

SOILS

	Unit Price
Sieve Analysis (ASTM D2217, D422 and D1140)	
a. Standard Sieves 3-inch through No. 200	\$130
b. No. 200 to .005 mm.(ASTM D422); add to a or c	130
c. Minus 3" to #4 or Minus No. 4 to passing No. 200	85
d. Percent passing No. 200 sieve	85
Percent Passing No. 200 Sieve and Liquid-Plastic Limit	195
Specific Gravity (ASTM D854)	145
Liquid-Plastic Limit (ASTM D4318)	145
Moisture-Density Determination	
ASTM D698 (Standard Proctor)	160
ASTM D1557 (Modified Proctor)	170
One Point Test	100
Moisture Content (ASTM D2216)	10
R Value (ASTM D2844)	400
CBR Test (ASTM D1883) - Not Including Proctor	400
Relative Density (ASTM D4253 & D4254)	275

AGGREGATES

Sieve Analysis (ASTM C136 & C117)	
a. Coarse aggregate Above No. 4 (ASTM C136)	\$65
b. Fine Aggregate No. 4 through No. 200	65
c. Coarse & Fine Aggregates through No. 200	130
d. Large Pit-Run Samples (+3 in.), per hour	85
Specific Gravity & Absorption, Coarse or Fine (ASTM C127 & C128)	130
Unit Weight (ASTM C29)	80
Los Angeles Abrasion (ASTM C131 or C535)	250
Sodium or Magnesium Sulfate Soundness (ASTM C88) Coarse or Fine; 5 cycles	350
Organic Impurities (ASTM C40)	80
Clay Lumps and Friable Particles (ASTM C142)	130
Flat and Elongates (ASTM D4791)	125
Lightweight Particles Coarse or Fine, SpG 2.0 & 2.4 (ASTM C123)	350
Fractured Faces (ASTM D5821 or applicable DOT std.)	125
Alkali Silica Reactivity 14 Day Test (ASTM C1260 or C1567)	
Coarse or Coarse/Fine Blend Aggregate	1000
Fine Aggregate	950
Alkali Silica Reactivity 28 Day Test (ASTM C1260 or C1567)	
Coarse or Coarse/Fine Blend Aggregate	1,300
Fine Aggregate	1,200
Fine Aggregate Angularity, (ASTM C1252, Method C)	125
(Aggregate Specific Gravity Test is Not Included)	
Sand-Salt Mixture Salt Content (CP-L 4216)	250
Sand Equivalent (ASTM D4318)	125

DIRECT CHARGES

Per Diem and Lodging	\$65/day + cost of lodging + 15%
Automobile or Pick-up	\$100/day min. or \$1.50/mile
Heavy-Duty Truck	\$110/day min. or \$1.75/mile
All other direct project expenses, i.e. special equipment rental, Commercial travel, bulk reproduction, protective clothing, sample shipping, etc.	Cost + 15%
Provide on-site field laboratory facilities and test equipment	Quotation on Request
GEOTECHNICAL AND ENVIRONMENTAL ENGINEERING SERVICES	Fee Schedule Available on Request

PORTLAND CEMENT CONCRETE AND MASONRY

	Unit Price
Vapor Emission Test (ASTM F1869/F2170), per location	\$135/150
Compressive Strength Testing	
a. Concrete Cylinder (ASTM C39) 4" x 8"	\$25
b. Concrete Cylinder (ASTM C39) 6" x 12"	\$30
c. Grout/Mortar 2" X 2" Cubes (IBC or ASTM C109)	20
d. Grout Prism (ASTM C1019)	45
e. Concrete Core (ASTM C42)	70
f. Flexural Strength (ASTM C78)	85
g. Splitting Tensile Strength (ASTM C496)	85
Cylinder Molds (4" x 8") per case	40
Cylinder Molds (6" x 12") per case	65
Masonry block/prism Assemblage Compressive Strength	
a. 8" unit	\$180
b. 12" unit	\$200
Unit Weight of Cylinder or Core (ASTM C567)	60
Concrete Mix Design (Aggregate Tests & Travel Not Included)	
a. Compressive (lab batched)	\$1,800
b. Compressive (truck batched)	1,500
c. Flexural (truck batch/lab tested)	2,000
d. Flexural (lab batch/lab tested)	4,000
Coring Concrete (ASTM C42), Two Man Crew, per hour	170
Dry Shrinkage (ASTM C157)	700

HOT MIX ASPHALT (HMA)

Extraction of Asphalt (ASTM D6307)	\$120
Extraction and Gradation (ASTM D6307, D5444)	240
Marshall unit weight, stability & flow, 3 specimens, pre-mixed (ASTM D6926-6927)	250
Coring Hot Mix Asphalt (ASTM D979)	
4" & 6" diameter cores, Two Man Crew per hour	170
Bulk Specific Gravity & Thickness (ASTM D2726)	50
HMA Trial Mix Design: (Aggregate Tests Not Included)	
Marshall Method (4 bin mix max.)	4,000
Superpave SHRP Gyratory (4 bin mix max.)	4,500
Additional mix design bin/aggregate	500
Theoretical Maximum Specific Gravity/Rice (ASTM D2041) per point	120
Immersion-Compression or Lottman (ASTM D1074 & D1075/ AASHTO T283)	600
Mix Design Volumetrics 1-point Verification (pre-mixed)	600
Mix Design Volumetrics 1-point Verification (lab-mixed)	1,200
Mix Design Update (Aggregate Verification Only); includes gradation and specific gravity of 4 materials maximum)	1,200
Tests not shown above	Quotation on Request

"Rush laboratory service & less than 24-hour notice for field testing available at 50 percent surcharge based upon availability"